Giving ChatTFT Direction: Decision-First Investigation Protocol, Not a Skill Tree

ChatTFT's missing "direction" is best fixed by organizing guidance around the decision and the contrast that would change it, not around a topic hierarchy of skills: a short, always-loaded investigation protocol (decision → alternatives → contrast → confounder check → threshold → stop), a small set of composite tools that encode the contrasts good players actually make, thin model-selected "cards" for data limits and statistical lenses, and a curated case library — with a skill hierarchy demoted to a lightweight index rather than the organizing principle.

TL;DR

- **The weakness is a framing problem, not a tool-selection problem.** A valuable answer differs from a correct lookup by (1) conditioning on the player's actual situation, (2) choosing the right baseline (within-shell, same-cost, same-holder), (3) auditing the obvious confounder, (4) converting deltas into a decision threshold, and (5) stopping when no remaining query could flip the recommendation. Topic-organized skills (itemization, comps, units) encode entities, but workflows are shaped by decision type and evidence availability, which cut across entities.
- **Recommended hybrid:** an always-on ~40-line investigation protocol + 3–4 composite contrast tools (shell-variant compare, within-holder item compare, component-constrained build ranking, confounder audit) + ~10 thin cards (data-limit, method-lens, entity) + ~15 worked cases retrieved by decision shape, with an answer-rubric critic only on the deep tier. This is a synthesis of Anthropic's "right altitude"/progressive-disclosure guidance, Agent Workflow Memory's evidence that induced workflows help, and evidence that instruction and tool bloat degrade performance.
- **Build the smallest version first and test it against a budget-matched baseline:** the protocol + two composite tools + data-limit cards, evaluated pairwise against the current system and a "manual lookup" baseline on ~150 logged questions, with expert-validated LLM judging and a split-half replication check on every quantitative claim. The key assumptions to test are that players value conditional recommendations over tables, that within-shell contrasts are stable enough to act on, and that the added calls pay for themselves.

Key Findings

Established (from primary sources):

- Anthropic distinguishes workflows (predefined code paths) from agents (LLMs directing their own process) and names five composable patterns — prompt chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer — recommending the simplest

pattern that works. aihero Evaluator-optimizer is described as effective "when we have clear evaluation criteria, and when iterative refinement provides measurable value." studocu

- Anthropic's context-engineering guidance says system prompts should sit at the "right altitude": between brittle hardcoded if-else logic and vague guidance that "falsely assumes shared context," giving "strong heuristics" instead. anthropic + 2 It recommends "diverse, canonical examples" over exhaustive rule lists.
- Agent Skills use three-level progressive disclosure: name/description always loaded, the SKILL.md body loaded when relevant, and bundled files loaded only as needed; Anthropic frames a skill as "an onboarding guide for a new hire." anthropic
- Instruction density degrades adherence: in IFScale, "even the best frontier models achieved only 68% accuracy" at 500 simultaneous instructions, alphaxiv with primacy effects (earlier instructions favored). emergentmind Arize's May 2026 re-run reports that the point where frontier models lose track has moved from roughly 200–300 simultaneous constraints to "closer to 2,000 instructions" depending on model, with current models "still perfect at 500," so the absolute numbers are moving fast, though the direction (more instructions → more omissions) still holds.
- Tool-set bloat degrades tool selection: RAG-MCP reports baseline tool-selection accuracy of 13.62% vs 43.13% with retrieval-filtered tools and >50% prompt-token reduction; alphaxiv OSWorld-MCP found removing tool filtering (158 tools) cut accuracy on tool-beneficial tasks from 24.9 to 18.0. arxiv
- Reusable workflows induced from past trajectories help: Agent Workflow Memory (Wang et al., 2024) reports 24.6% and 51.1% relative success-rate improvements on Mind2Web and WebArena. github But Hajimiri et al. (arXiv 2606.15017, 2026) found that across four WebArena domains and three models "the vanilla baseline matches or surpasses all three augmentation methods in aggregate success rate while often using fewer total tokens" (e.g., Qwen 3.6-27B: budget-matched vanilla 42.14% at 95.5K tokens vs AWM 38.01% at 115.0K) — memory/skill modules are not automatically worth their tokens.
- As-needed decomposition beats fixed upfront plans in interactive tasks: ADaPT decomposes only when the executor fails, reporting success rates up to 28.3%, 27%, and 33% higher than baselines on ALFWorld, WebShop and TextCraft. aclanthology linkedin
- Multi-step analysis agents outperform single-query agents on insight generation: InsightBench evaluates agents on "formulating questions, interpreting answers, and generating a summary of insights and actionable steps," and its multi-step AgentPoirot beat a single-query Pandas Agent. github
- Spending more compute explains much of multi-agent advantage: Anthropic's research system beat single-agent Opus 4 by 90.2% on an internal eval, backfromthefuture but

"token usage by itself explains 80% of the variance," and, per Anthropic, "multi-agent systems use about 15× more tokens than chats" (agents alone about 4×).

- Cost must be part of evaluation: Kapoor et al. ("AI Agents That Matter") show simple baselines can match complex agents on HumanEval at far lower cost and argue all agent evaluation should be cost-controlled and plotted on an accuracy–cost Pareto frontier. x normaltech
- LLM judges approximate human preferences with caveats: GPT-4 judges reached "over 80% agreement" with humans on MT-Bench, matching human–human agreement, but show position, verbosity, and self-enhancement biases. arxiv
- Expert decision-makers mostly recognize the situation and evaluate one option by mental simulation rather than comparing all options (Klein's Recognition-Primed Decision model); in Klein, Calderwood & Clinton-Cirocco's study of 26 fireground commanders across 156 decision points, "in over 80% of the decision points" commanders identified the situation as typical of a standard prototype, and fewer than 12% showed any simultaneous comparison of options. Pirolli & Card's analyst model separates a foraging loop (find/filter evidence) from a sensemaking loop (schema, hypotheses, evidence marshalling, decision).
- TFT-specific bias patterns are documented by practitioners: ElbroC shows 5-cost survivorship (Ahri's 3.75 average placement looked dominant but ranked second-worst among 5-costs once compared within cost), omitted-variable bias (excluding Piltover "drops the avg. of Jayce by 0.35"), item deltas that must be re-read inside the comp (Kai'Sa + Morello at a −0.32 global delta), rank-filter shifts (+0.40 between Master+ and Gold+), and tiny samples (an augment at 4.10 on 21 games). Substack Substack
- Riot actively shapes what stats exist: Mortdog announced on X in November 2024 that Riot would "remove Augments from match history, starting with the launch of Into the Arcane" (Set 13), because "players rush to conclusions based on that data"; an earlier 2023 removal was reverted. Riot's developer policy states that products "should not remove game decisions" and should "protect the learning curve." riotgames The live set is Set 18 (Enchanted Wilds, launched Aug 26, 2026) with patches roughly every two weeks and sets roughly every four months. seemeta tftsense

My proposals/inferences (not established): everything in the architecture, question classification, composite tool designs, stopping rules as written, and the evaluation design below. Where a choice rests on weak evidence, I say so.

Details

1. Reframing the problem

The current system's prompts are mostly entity-and-method knowledge: how to resolve names, which ranking projection to use, what sample tiers mean, which delta tools exist. That

is why tool use is good and direction is weak. What's missing is a representation of the decision the player faces, which determines:

- **The comparison set** (alternatives the player can actually choose between — not "all items," but "items buildable from Bow + Rod + Belt that fit this holder").
- **The right population** (Jinx boards where Jinx is the carry with 3 items — not all Jinx boards).
- **The baseline** (same cost tier, same shell, same level).
- **The threshold** (what size of difference should change behavior given CI width and plausible confounding).
- **The stopping point** (when the recommendation can no longer flip).

A skill hierarchy organized by topic ("itemization > radiant items", "comps > pivots") routes on nouns in the question. But two questions with the same noun — "best items on Jinx" and "I have Bow/Rod/Belt, is Jinx worth it?" — require different workflows, and two questions with different nouns — "is Guinsoo's or Red Buff better on X" and "is comp A or comp B better at level 8" — share the same workflow (a controlled pairwise contrast). This is the central reason a topic hierarchy underperforms: **workflow shape is determined by decision type × evidence availability, which is orthogonal to entity type.** This is my inference, but it is consistent with Pirolli & Card's finding that analyst leverage points concentrate in "problem structuring … evidentiary reasoning … and decision making," not in information categories. researchgate

2. Question bank (36 realistic questions)

Tiers: **L** = lookup (1–2 calls), **S** = standard contrast (3–6), **D** = deep (6–12), **U** = unsupported/partially supported (answer the answerable part, name the missing evidence, give an in-game decision rule).

#	Player question	Decision supported	Tier	Evidence with these tools	"Merely correct" vs "valuable"
1	Best items on Jinx?	What to build on a known carry	L	rank_items holder=Jinx; rank_unit_loadouts	Table of items vs top 3-item loadout + which item is the flexible slot
2	Most played 4-costs?	Meta awareness	L	rank_units cost=4 sort=games	Fine as a list; add pick-rate vs placement tension in one line
3	Strongest 4-costs?	What to aim for	S	rank_units cost=4 by avg; control star level	Raw avg vs within-star (2★) comparison; flag 3★ inflation
4	Guinsoo's or Red Buff on X?	Slam choice	S	compare_cohorts within X holders with 3 items	Global item ranks vs holder-matched contrast with CI, "choose by components if within CI"
5	I have Bow, Rod, Belt and a Shojin emblem — what should I build?	Component allocation	D	rank_items constrained; loadouts; emblem holder deltas	Generic best items vs ranked buildable plans incl. emblem holder
6	Is comp X still good without the 5-cost cap?	Commit/pivot	D	compare_cohorts shell with vs without cap, level controlled	"X averages 4.1" vs "without cap at level 8 it's 4.4; cap worth ~0.4; don't force"
7	What's overrated right now?	Avoid traps	D	high games, weak delta vs cost/star baseline	Low-avg list vs popular-but-underperforming after controls
8	Why does Y have good placement but low play rate?	Interpret a stat	S	Y cohort by level/star; co-occurring traits	"It's strong" vs "it's a late-game cap; conditional on level 9"

#	Player question	Decision supported	Tier	Evidence with these tools	"Merely correct" vs "valuable"
9	Which augment for comp X?	Augment pick	U	none (no augment data)	Refuse vs answer trait/emblem proxies + in-game rule
10	Should I roll at 4-1 or level?	Tempo	U/partial	rolldown_probabilities; no path data	Odds only vs odds + what HP/econ would flip it
11	Chance to hit Jinx 3★ with 50 gold at 7?	Roll budget	L	rolldown_probabilities	Probability + "if you hold N copies…" sensitivity
12	Who should hold the Radiant item?	Item allocation	S	rank_items Radiant group_by_holder	List vs holder comparison within shell
13	Is the X emblem worth slamming?	Emblem value	S	trait tier deltas with/without emblem holder	Emblem avg vs incremental tier breakpoint value
14	Best 2-item start for X before 3rd item?	Mid-game build	S	loadouts item_count=2	Hard: final-board only; note survivorship
15	What's the best comp?	Commit target	S	rank_traits / cohort by trait set	Tier list vs "by your level/flex"
16	Is Bard good in reroll?	Comp viability	S	Bard star3 cohort	Avg vs vs non-reroll baseline
17	Which tank items on frontline in X?	Itemization	S	item_deltas in shell for tank holders	Global vs in-shell
18	Is 3-star Y worth it vs leveling?	Reroll decision	D/partial	Y 2★ vs 3★ at levels 7/8	Placement gap + rolldown odds of 3★
19	What to do with spare Belt + Chain?	Component use	S	rank_items buildable among holders	List vs which holder in your shell

#	Player question	Decision supported	Tier	Evidence with these tools	"Merely correct" vs "valuable"
20	Is 6 X trait better than 4 X + splash?	Trait breakpoint	S	rank_traits tier compare within shell	Avg vs controlled tier contrast
21	Does Artifact Z beat a completed item?	Artifact value	D	holder-matched; confounded by artifact-granting augments	Must flag augment confound
22	Which carry for this shell: A or B?	Carry selection	S	compare_cohorts shell+A vs shell+B	Clean pairwise
23	What do top players do in EUW vs NA?	Regional meta	S	group_by region	Partial: region ≠ skill
24	Is X better in high elo?	Skill-conditional	U	no rank data	Name gap
25	How to position vs assassins?	Positioning	U	none	Principle + no data
26	Was X nerfed hard last patch?	Patch trend	U	single patch only	Current-patch only
27	Why did I go 6th with this board?	Player review	U	no match history	Compare board to winning variants; can't see path
28	Is it contested-proof?	Contest risk	U	no lobby data	Flex alternatives within shared units
29	Best flex line from Ahri opener?	Pivot options	D	shells sharing units; partial	Shells containing unit + transitions unseen
30	Should I hold or slam Spatula?	Hold/slam	U/partial	emblem outcomes, no timing	Outcome of emblem lines, timing unknown

#	Player question	Decision supported	Tier	Evidence with these tools	"Merely correct" vs "valuable"
31	Is level 9 cap worth it for X?	Level target	S	X cohort group_by level	Level effect confounded with strength
32	Anything sleeper?	Discovery	D	low play, strong controlled delta, n≥500	Hidden gems with stability check
33	Is Rageblade core or replaceable?	Rigidity	S	loadout with/without	Rigidity score
34	Do Support items matter in X?	Item family	S	item_type Support deltas in shell	
35	What if I only have 2 items for carry?	Constrained board	S	unit_item_count=2 cohort	Realistic expectation
36	Which 5-cost is best?	Cap choice	S	same-cost, level-controlled	Survivorship correction

Distribution check: roughly 4 L, 19 S, 7 D, 9 U/partial — a design built only for deep analysis would over-spend on most traffic, and one built only for lookups misses the highest-value third.

3. Approaches compared

Approach	Player value	Flexibility	Reliability	Impl. complexity	Latency/cost	Maintainability (set churn)
(a) Skill hierarchy (topic tree, model-selected)	Medium: good recipes for covered topics	Low–medium: misroutes cross-cutting questions	Medium: prompt pulls loosely related skills	Medium	Low–medium	Poor if skills embed set content
(b) Decision-archetype router + playbooks	Medium–high on covered archetypes	Low: rigid on hybrids	High within archetype	Low–medium	Low	Good: archetypes are set-agnostic
(c) Single investigation protocol + thin reference cards	High if model applies it well	High	Medium: depends on model judgment	Low	Low–medium	Very good
(d) Planner–executor–critic / evaluator-optimizer	Medium–high on deep questions	High	Higher on deep tier	Medium–high	High (2–3× calls)	Good
(e) Faceted modules (decision, entity, lens, data-limit, format)	High	High	Medium–high	Medium	Low–medium	Good if facets are orthogonal
(f) Case library of worked investigations	High (shows judgment)	Medium	Medium–high	Medium	Low	Cases decay per set; must re-curate

Approach	Player value	Flexibility	Reliability	Impl. complexity	Latency/cost	Maintainability (set churn)
(g) Composite contrast tools	High: encodes correct baseline	Medium	High: removes error-prone steps	Medium–high (backend)	Lowest per question	Very good: set-agnostic logic
(h) Hybrid: c + g + e-cards + f + d-on-deep	**Highest expected**	High	High	Medium	Tiered	Good

Why not (a) as the backbone: the existing `tft-unit-itemization` skill is a symptom — it prescribes calling all three delta tools for every itemization question, which is the topic-tree failure mode (procedure fixed by topic, not by what the decision needs). Hierarchies also force selection among overlapping siblings; the RAG-MCP and OSWorld-MCP results on tool overload suggest selection among many similar options is itself a failure point, github though those studies measure tools, not skills — the transfer to skill selection is my inference.

Why not (d) everywhere: Anthropic's own numbers show multi-agent gains largely track token spend, 13labs and the budget-constrained WebArena study found extra modules lost to a budget-matched simple actor. arxiv A critic is worth it only where the answer quality rubric is clear and the question is deep.

Why (g) matters most mechanistically: the single most common analytic error in this data is the wrong baseline (global instead of within-holder/within-shell). Encoding the baseline in a tool moves that judgment from a prompt instruction the model may drop (IFScale-style omission) into code that always runs.

4. Recommended architecture

Layer 1 — Always-on investigation protocol (system prompt for main + data_analyst). Sketch:

```
Before any tool call, write a 3-line frame:
  DECISION: what the player will do differently based on this answer
            (e.g., "slam Guinsoo vs hold for Red Buff on X").
  OPTIONS: the 2–5 choices actually available to them given what they told you.
  PIVOTAL CONTRAST: the single comparison that would most change the choice,
            with its population and baseline (same holder / same shell /
            same cost / same level).
Then choose a tier:
  LOOKUP (one call answers the decision), CONTRAST (3–6 calls),
  DEEP (needs confounder audit or multiple contrasts), or
  OUT-OF-DATA (decision hinges on augments, path, contest, rank, patches,
  player history). Say which, briefly.
Investigate:
  - Run the pivotal contrast first. Read n, CI, overlap.
  - Run ONE confounder check only if the leading explanation has an
    obvious rival (star level, level, 3-item highroll, cost survivorship,
    artifact/emblem-granting augments, trait co-occurrence).
Stop when any is true:
  - the leading option's edge exceeds CI width and plausible confounding;
  - no remaining query could flip the recommendation;
  - the remaining uncertainty lives in data you don't have.
Answer:
  - Lead with the recommendation for THEIR situation.
  - State the contrast in placement deltas ("~0.3 better than the next
    buildable option, same holder, n≈2,000").
  - One line on what would change the answer and the in-game signal to watch.
  - Omit tables unless asked.
```

This is ~25 lines — deliberately short to stay at the "right altitude" and well below the instruction densities where IFScale shows omissions.

Layer 2 — Composite contrast tools (backend). Proposed signatures:

- `compare_shell_variants(shell: FilterGroup, variants: [FilterGroup], control_level?: bool)` → each variant vs shell baseline with lobby-clustered CIs, level-stratified. Covers "without the cap," "A vs B carry," "6 vs 4+splash."
- `compare_items_on_holder(unit, items: [..], holder_item_count=3, star?, shell?)` → holder-matched item contrasts with highroll control (fixed item count, fixed star).
- `rank_buildable(components: [..], emblems/artifacts: [..], shell?: FilterGroup)` → ranks complete plans (which items, on which holders) buildable from the stated components, using loadout data.

- `confounder_audit(cohort, effect_dim)` → re-runs the effect stratified by star level, level, item count, and top co-occurring trait; reports whether the effect survives each stratification.

Layer 3 — Thin cards (model-selected, progressive disclosure, each ≤30 lines).

- Data-limit cards: augments, path/tempo, contest, rank, cross-patch, player history, positioning. Each card states what is missing, the closest answerable proxy, and a generic in-game decision rule. These are the most set-stable content.
- Method-lens cards: survivorship (cost/level), highroll (3-item, 3-star), confounding (traits, augment-granted artifacts/emblems), small samples, frequency vs outcome, delta vs relative_delta.
- Entity cards: item families (Radiant, Artifact, Emblem, Support), trait style tiers, holder archetypes. These replace the current itemization skill's reference content without its fixed procedure.

Layer 4 — Case library (~15 cases initially). Each case is a short worked investigation keyed by decision shape (constrained allocation, pairwise slam, shell variant, trap detection, stat explanation, out-of-data). Retrieve 1–2 by decision shape, not by entity. Later, induce new cases from logged high-rated sessions, AWM-style — but only after the budget-matched test shows they earn their tokens.

Layer 5 — Critic on the deep tier only. A single rubric pass: does the answer name the decision, the baseline, the confounder check, the threshold, and what would change it? Revise once if any is missing.

Where the skill hierarchy survives: as the index over cards and cases (names + one-line descriptions, which is exactly Skills' level-1 metadata), not as the planner.

5. Worked examples (unit names illustrative; numbers hypothetical, not real Set 18 data)

Example 1 — "Best items on Jinx?" (LOOKUP). Frame: decision = what to build on a committed carry; options = top items; pivotal contrast = top 3-item loadouts on Jinx. One call: `rank_unit_loadouts(unit=Jinx, item_count=3)`. Stop: one call answers it. Answer: "Default build is Guinsoo + IE + Last Whisper; IE is the flexible slot — Giant Slayer is within noise if you lack Glove." Value over lookup: the flexible-slot note, from looking at which item varies across top loadouts.

Example 2 — "I have Bow, Rod, Belt and a Shojin emblem — what should I build?" (DEEP). Frame: decision = allocate components now; options = buildable items × plausible holders + emblem holder. Pivotal contrast: `rank_buildable(components=[Bow, Rod, Belt], emblems=[Shojin])`. Confounder check: emblem lines co-occur with the trait breakpoint — check emblem value at fixed trait tier. Answer: "Build Guinsoo (Bow+Rod) for your carry, keep Belt;

put the emblem on your 4-cost frontline to hit the next breakpoint — that breakpoint, not the emblem holder, accounts for most of the edge."

Example 3 — "Is comp X still good without the 5-cost cap?" (DEEP). Frame: decision = commit to X at level 8 or pivot. Pivotal contrast: `compare_shell_variants(shell=X core, variants=[with cap, without cap], control_level=true)`. Confounder: level (cap boards are level 9). Answer: "At level 8 without the cap, X places ~4.4 vs ~4.0 with it at level 9; comparing at the same level, the cap is worth ~0.2. X is fine to play at 8 if you're healthy; don't roll everything for the cap."

Example 4 — "Guinsoo's or Red Buff on X?" (CONTRAST). Pivotal contrast: `compare_items_on_holder(unit=X, items=[Guinsoo, Red Buff], holder_item_count=3)`. Stop if the CI overlaps: "Within noise — slam whichever your components make now; Guinsoo has the more consistent top-4."

Example 5 — "What's overrated right now?" (DEEP). Frame: avoid traps; options = popular units/comps. Contrast: high-games units with deltas versus same-cost, same-star baseline. Confounder audit on the top 3 candidates. Answer: name 2–3 that are popular but underperform after controls, and say why (e.g., "only good at 3★, which most boards don't reach").

Example 6 — "Why does Y have good placement but low play rate?" (CONTRAST). Cohort by level and star. Answer: "Y is mostly played at level 9 by already-winning boards — a survivorship artifact. Compared within level 9, it's average."

Example 7 — "Which augment for comp X?" (OUT-OF-DATA). Data-limit card: no augment data (Riot removed augments from match history starting Set 13). manifold Answer: "I can't rank augments from this data. What I can say is which trait breakpoints and emblems X converts best, so prioritize augments that give those. In-game: take the econ augment if you're above 60 HP."

Example 8 — "Should I roll at 4-1 or level?" (OUT-OF-DATA, partial). `rolldown_probabilities` gives odds; no path data. Answer: "With 50g at 7 you hit 2★ carry ~70%; leveling to 8 drops it to ~55%. The real deciding variable is HP and streak, which I can't see — roll if below ~50 HP."

6. Answers to the five questions

What distinguishes a valuable workflow from technically correct tool calls? A valuable workflow is anchored on a decision and its alternatives, runs the pivotal contrast first with the correct baseline, checks the one confounder that could reverse it, expresses results as placement deltas relative to the player's options, and stops when the answer cannot flip. Technically correct sequences answer the literal noun of the question with an unconditioned ranking.

What is reusable vs question-specific? Reusable: the protocol, the baselines (same holder/shell/cost/level), the confounder catalog, the stopping rules, the data-limit cards, the answer shape. Question-specific: choosing the pivotal contrast, judging which confounder is plausible, deciding the threshold given the player's situation, and translating to in-game advice.

How to organize/select/combine guidance? One always-on protocol, composite tools for the common contrasts, thin orthogonal cards selected by the model, and 1–2 retrieved cases keyed by decision shape. Keep the total loaded guidance small; demote the hierarchy to an index.

What instructions help prioritize, interpret, and answer? Write the decision frame before calling tools; name the pivotal contrast; read n/CI/overlap before interpreting; run one confounder check; lead with the recommendation; quantify in deltas; state what would change the answer.

How to evaluate value vs time/complexity? See below.

7. Evaluation plan

- **Question set:** ~150 questions from logs, stratified by tier (L/S/D/U) and decision type; include the 36 above as seeds; hold out 30% for final comparison.
- **Baselines:** current system; a "manual lookup" baseline (one top-level ranking call + plain summary, simulating a stats-site lookup); a budget-matched baseline (current system allowed the same number of calls/tokens as the new one).
- **Rubric (1–5 each):** decision relevance; value over lookup (would the player act differently than with the lookup?); statistical correctness (baseline, population, confounder); calibration (confidence matches evidence; limits named); efficiency (calls, latency).
- **Pairwise preference:** randomized order, both orders judged (to counter position bias), length-normalized instructions to counter verbosity bias. wandb
- **Judge validation:** 3+ high-ladder players rate a 50-question subset; require judge–expert agreement near human–human levels before trusting judges at scale.
- **Claim replication:** split boards into two halves (by region or time); every quantitative claim made on one half is checked on the other; report replication rate.
- **Cost/latency budgets per tier:** L ⩽2 calls, S ⩽6, D ⩽12; report accuracy–cost Pareto curves.
- **Ablations:** protocol only; +composite tools; +cards; +cases; +critic.

8. Practical starting point

1. Write the protocol (~25 lines) into main and data_analyst prompts.
2. Build `compare_items_on_holder` and `compare_shell_variants`.
3. Write the 7 data-limit cards.
4. Run the pairwise eval on ~50 questions vs current and the budget-matched baseline.

5. Only then add cases, the confounder audit tool, and the critic.

9. Assumptions to test before expanding

- Players prefer conditional recommendations over tables (test with expert ratings and in-product thumbs).
- Within-shell contrasts are stable enough to act on (test with split-half replication).
- The extra calls produce value beyond a budget-matched baseline (test with the ablation).
- Decision-shape case retrieval beats entity retrieval (test head-to-head).
- The model reliably writes a useful decision frame (audit 100 frames by hand).

Caveats

- Several research findings (tool overload, instruction density) were measured on tools or keyword constraints, not on analytic skills; transfer to skill selection is inference.
- The worked examples use illustrative units and hypothetical numbers.
- Practitioner sourcing on TFT stat misuse is thin: ElbroC's article is the main written source; I did not find verbatim statements from Dishsoap, Frodan, or Mortdog on average placement vs top-4 or highroll skew.
- Augment data status in late 2026 is unconfirmed; the current tools lack augments regardless. manifold
- Set churn (~4-month sets, ~2-week patches) means any set-specific content in cards or cases decays quickly; keep set-specific content out of the protocol and tools.